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Tensile deformation behavior analysis of low density Fe–18Mn–10Al–*x*C steels

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ABSTRACT

Three low density Fe–18Mn–10Al–*x*C steels containing 0.5, 0.8 and 1.2 C (wt%) were utilized to investigate the differences in microstructures and their influence on mechanical behaviors during plastic deformation. The 0.5C steel had a duplex ferritic–austenitic structure and the fraction of austenite was ~59.2% while the fraction of austenite was ~84.8% for the 0.8C steel and some ordered phases existed in both ferrite and austenite. In contrast, a fully austenitic microstructure with some κ carbides was achieved in the 1.2C steel. Due to the existence of the ordered phases by the addition of C from 0.5% to 0.8%, the 0.8C steel exhibited a highest tensile strength of ~979 MPa, and a moderate strain hardening capacity. The 1.2C steel revealed a superior combination of strength and ductility (ultimate tensile strength of 946.7 MPa and elongation to failure of 56%), with a much more pronounced strain hardening than the other two steels. The differential Crussard–Jaoul (C–J) analysis demonstrated a two-stage strain hardening behavior in both 0.5C and 0.8C steels, while a three-stage one in the 1.2C steel. This difference in strain hardening behavior was further understood in terms of microstructural analysis at the different stages of plastic deformation.

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1. Introduction

The target of high performance steels is to increase strength and maintain ductility of steels as to satisfy the needs for weight savings by reducing the thickness of structural automotive components. However, the requirement for the stiffness of structures cannot be satisfied when the thickness of the structure is thinned to a certain extent. That is why lowering the density of steels has become one of the research trends in the development of high performance steels.

In the previous research, it was revealed that the density of Fe–Mn–C steels can be reduced to 6.5–7.1 g/cm^3 with an addition of 6.5–12% Al [1]. Generally, low density steels can be classified into three categories: (1) ferrite based steels [2,3]; (2) austenite based steels [4–6] and (3) duplex or triplex steels [1,7–12]. The physical phenomena in duplex and triplex low density steels are quite complex due to the different deformation mechanisms in each phase and the interactions among the major phases and the ordered ones. Frommeyer et al. [1] investigated the high-strength Fe–(18–28)Mn–(9–12)Al–(0.7–1.2) C triplex steels exhibiting a strength of 700–1100 MPa and a total elongation up to 60% or more. Accordingly, the strain hardening exponents were determined to investigate the tensile deformation behaviors of the steels. Later, Sutou et al. [7] demonstrated that an addition of C to Fe–20Mn–10Al–*x*C alloys, ranging from 0.25 to 1.5% linearly increased the tensile strength, with a maximum of ~10921 MPa. However, the tensile ductility of the steels first increased, reached a maximum value (~59.1%) at 1.0% C and then declined again (only ~17.1%) at 1.5% C. The influence of cooling patterns on the working hardening curves of the steels was studied as well, though the microstructural analysis during deformation was not provided in this study. Hwang et al. [8] studied a duplex Fe–20Mn–9Al–0.6C steel exhibiting a tensile strength of ~800 MPa and an elongation of ~46%. It was pointed out that the strain hardening exponent of austenite was higher than that of ferrite. Recently, the strain hardening behavior of a duplex steel with the similar composition, Fe–18.1Mn–9.6Al–0.65C, was investigated and multi-stage strain hardening characteristics were reported [9]. Another work [10] demonstrated that the Fe–27Mn–12Al–0.8C steel contained ordered phases in both phases, B2 and DO_3 in ferrite and κ carbide in austenite, in which a relatively high yield strength of 812 MPa and tensile strength of 955 MPa were achieved, with a moderate elongation of 42% due to a relatively low initial strain hardening rate. Meanwhile, our preliminary work [12] indicated that an addition of C ranging from 0.8 to 1.2% to the Fe–18Mn–10Al–*x*C steels caused a dramatic change in microstructures and hence in the final mechanical properties of the steels.

Although some studies have been conducted on the microstructure–mechanical properties relationship of such low density

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Fe–Mn–Al–C steels, the strain hardening behavior of duplex Fe–Mn–Al–C alloy system, especially those containing ordered phases has not been fully understood yet. It is well known that the Crussard–Jaoul (C–J) analysis (i.e. $\ln(d\sigma/d\varepsilon)$ vs. $\ln\varepsilon$) is considered as a useful tool to indicate the relative changes in work hardening behaviors over a range of plastic strains, especially in materials that exhibit microstructural evolutions (e.g. mechanical twinning or martensitic transformation) upon loading. Meanwhile, as a strong austenite stabilizer, an addition of C to high Mn and high Al alloys can significantly increase the volume fraction of austenite and also lead to the formation of ordered phases. In the present work, therefore, a Fe–18Mn–10Al–*x*C alloy system containing three different C contents was chosen to investigate the effects of the microstructures on the mechanical behavior of the steels. A particular emphasis will be placed on the clarification of the differences in mechanical behaviors in the steels consisting of different phase constituents by C–J analysis, which can provide some useful data for alloy design and applications of low density Fe–Mn–Al–C steels.

2. Experimental

Three different Fe–18Mn–10Al–*x*C (0.5, 0.8 and 1.2C, wt%) steels were selected, and the chemical compositions of the steels are shown in Table 1. The densities of three steels were calculated [7] as 6.92, 6.88 and 6.78 g/cm^3, respectively. The stacking fault energy (SFE) values of austenite in 0.5C and 0.8C steels are 73.4 and 77.9 mJ/m^2 and that of 1.2C steel is 84.5 mJ/m^2 according to the model we used previously [13]. The 50 kg ingots of steels were prepared by induction melting under an argon atmosphere. The ingots were forged at 1200 °C and then hot rolled after homogenization at the same temperature for 3 h to the thickness of 3.5 mm with a reduction of 90%. The hot-rolled strips were annealed at 1000 °C for 1 h and then cold rolled to a thickness of 1 mm. Then, the cold rolled strips were annealed at 1000 °C followed by water quenching to room temperature.

Tensile tests were carried out at room temperature using a universal testing machine (SANSCMT5000) operating at a constant crosshead displacement velocity (equivalent to an initial strain rate of 1.33×10^{-3} s^{-1}). Optical microscopy specimens were prepared by standard mechanical polishing procedures and etching with a solution of 1 ml hydrochloric acid, 100 ml methanol and 4 g picric acid. The volume fractions of different phases were measured by image analysis technique. Thin foils for transmission electron microscopy (TEM) observations were prepared by twin-jet electropolishing by using 9% perchloric acid in methanol at −30 °C. TEM analysis was performed in a Tecnai G^{2}20 TEM at an acceleration voltage of 200 kV.

3. Results

3.1. Microstructures before tensile deformation

Fig. 1 shows the representative optical microstructures of the as-quenched steels, revealing that 0.5C and 0.8C steels have a duplex ferritic–austenitic microstructure, while 1.2C steel is fully austenitic with some annealing twins in grains. The volume fractions of austenite are determined to be 59.2% and 84.8% in 0.5C and 0.8C steels, respectively. Detailed microstructure analysis under TEM reveals that no precipitates exist in both austenite (Fig. 2(a)) and ferrite (Fig. 2(b)) phases of 0.5C steel. In comparison, the diffraction patterns (SADPs) reveal some weak extra spots between the fundamental reflections in 0.8C and 1.2C steels, indicating the existence of precipitation particles, shown in Fig. 2(c), (e) and (f). Further indexing of these SADPs confirms that 0.8C steel contains a κ carbide phase [$(Fe,Mn)_3AlC_x$] in austenite (Fig. 2(c)) [14], and some B2 domains with fine DO_3 particles in ferrite (Fig. 2(e))[10]; whereas 1.2C steel contains some κ carbides in the matrix of austenite (Fig. 2(f)). As both ferrite and austenite phases contain ordered phases in 0.8Al steel, the steel is referred to complex low density steel in the present work.

Table 1
Chemical compositions of the investigated steels (mass, %).

Steel	Al	Mn	C	Fe
0.5C	9.90	17.71	0.48	Bal.
0.8C	9.91	17.14	0.77	Bal.
1.2C	10.19	17.45	1.16	Bal.

3.2. Mechanical properties and flow behavior

The engineering stress–strain curves of the steels are plotted in Fig. 3. A summary of the tensile properties of the present low density steels with three different C contents are listed in Table 2. As is observed with increasing C content from 0.5% and 1.2%, the total elongation (TE) is found to increase, reaching a maximum of 56.2% at 1.2% C, whereas the ultimate tensile strength (UTS) first increases, reaching a maximum value of ~979 MPa at 0.8% C, and then drops slightly to ~947 MPa at 1.2% C. 1.2C steel shows a superior combination of strength and ductility, i.e. the UTS × TE value can reach as high as 53.2 GPa%, which is almost twice as high as that of 0.5C steel.

The variation of strain hardening rate, $d\sigma/d\varepsilon$, with true strain of the three experimental steels are shown in Fig. 4, indicating quite different strain hardening behaviors. Here, the C–J analysis has been conducted to investigate the difference in strain hardening behavior of the steels. It can be found in Fig. 5 that 0.5C and 0.8C steels exhibit a two-stage strain hardening, while 1.2C steel manifests a three-stage strain hardening. Accordingly, in order to further investigate the influence of the initial microstructures on their deformation mechanisms, the microstructural changes during plastic deformation have been examined using a series of interrupted tensile tests combined with TEM.

3.3. Microstructural evolution during tensile deformation

The microstructural evolution of 0.5C steel during deformation is shown in Fig. 6. In the early deformation stage (~0.02 true strain), the traces of planar slip can be readily observed in austenite (Fig. 6(a)) and some dislocation tangles are also visible in ferrite (Fig. 6(b)). As deformation proceeds to a true strain of 0.18, intensive dislocation slip appears (Fig. 6(c)), and Taylor lattice (Fig. 6(d)) and microbands (Fig. 6(e)) have developed in austenite; whereas dense dislocation tangles can be observed in ferrite (Fig. 6(f)).

Fig. 7 shows the microstructural changes at different deformation stages of 0.8C steels under tension. At the beginning of deformation ($\varepsilon=0.01$), traces of planar slip can be found in austenite (Fig. 7(a)), and some dislocations exhibit a typical paired superdislocation feature in ferrite (Fig. 7(b)). At a true strain of 0.18, microbands have formed (Fig. 7(c)) in austenite, while a distinct increase in the dislocation density appear in ferrite (Fig. 7(d)).

Fig. 8 represents the microstructure changes during tensile deformation of 1.2C steel. Similar to 0.5C and 0.8C steels, planar slip can be found at a true strain of 0.02 (Fig. 8(a)). However, Taylor lattice-like structure has been easily observed in some regions of austenite when strained to 0.07 (Fig. 8(b)). Well-developed microbands can be observed at strain of 0.3 (Fig. 8(c)), and the interaction between microbands occur at a strain as large as 0.4 (Fig. 8(d)).

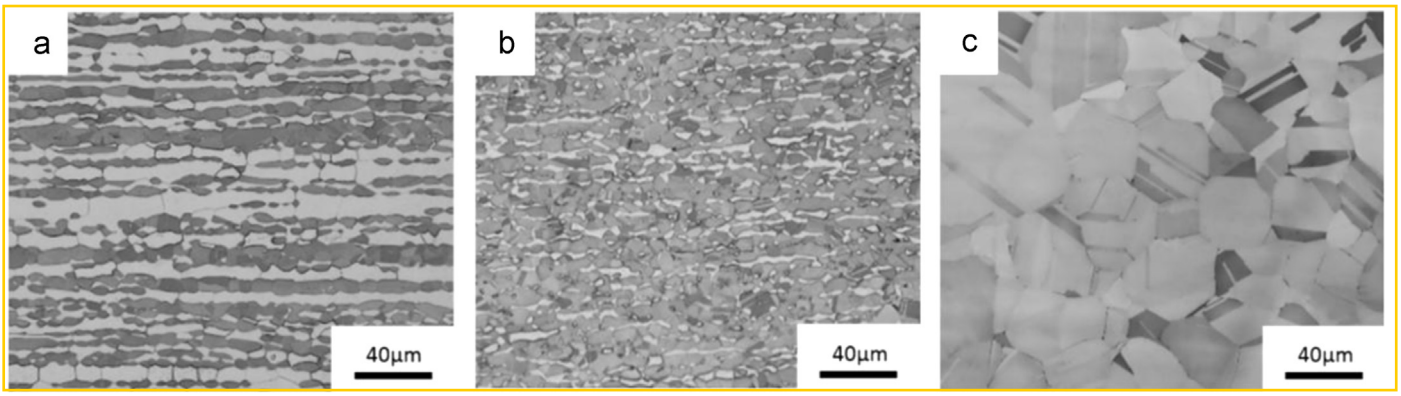

Fig.1. Optical microstructures of Fe–18Mn–10Al–*x*C steels (a) 0.5C steel, (b) 0.8C steel, (c) 1.2C steel.

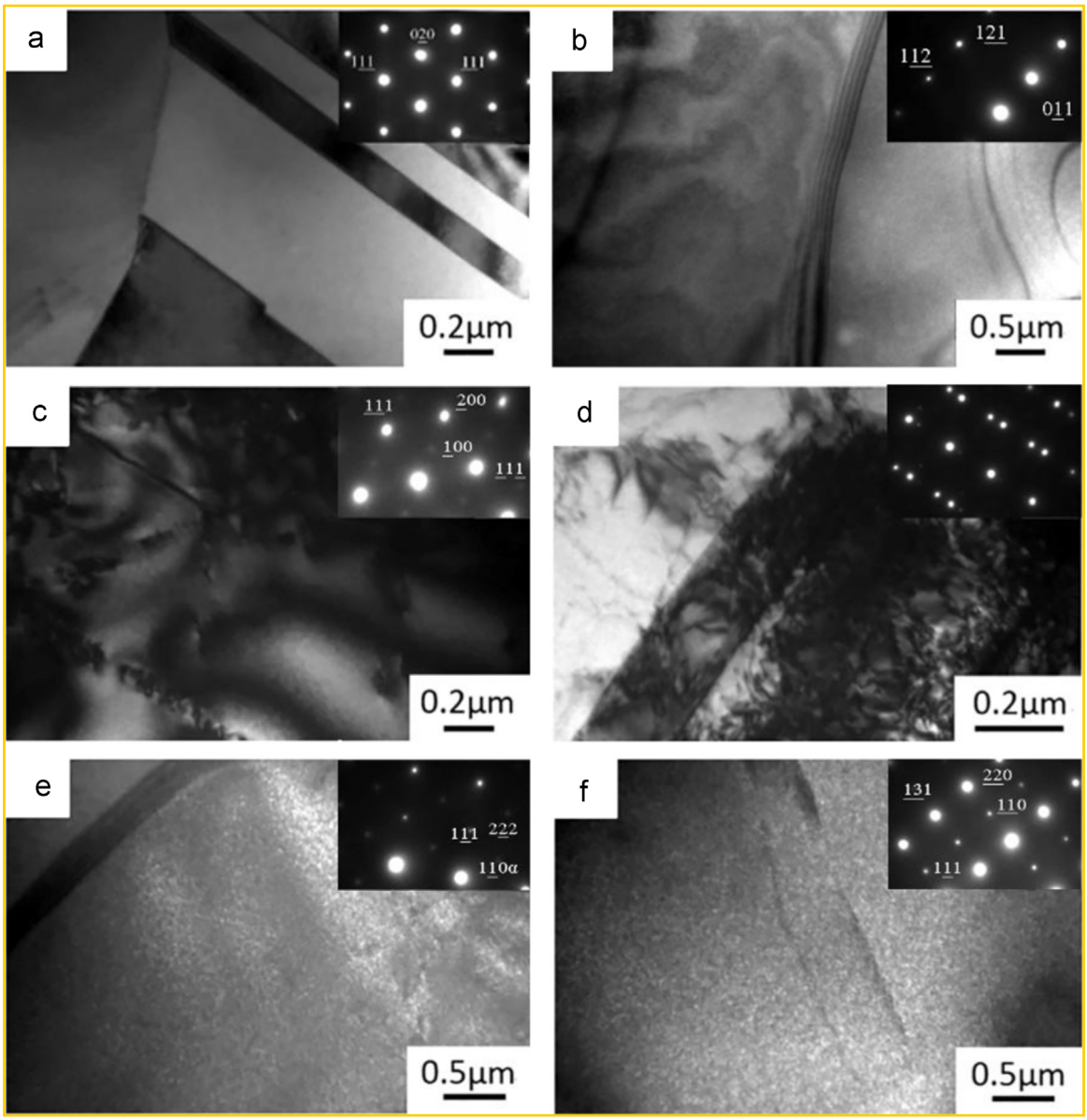

Fig. 2. Microstructures of Fe–18Mn–10Al–*x*C steels (TEM) (a)–(b) 0.5C steel: (a) γ phase, (b) α phase, (c)–(e): 0.8C steel (c) γ+κ (d) twins in γ phase, (e) dark field image of α phase, (f) dark field image of γ phase in 1.2C steel.

4. Discussions

There are several factors influencing the mechanical properties of the duplex, triplex or complex high Mn steels such as alloying elements, phase fraction, grain size as well as characteristics of each constituent phase. In Fe–Mn–Al–C system, Mn and C are austenite stabilizers and Al is a ferrite stabilizer. Meanwhile, various ordered phases appear in this system due to the heavy

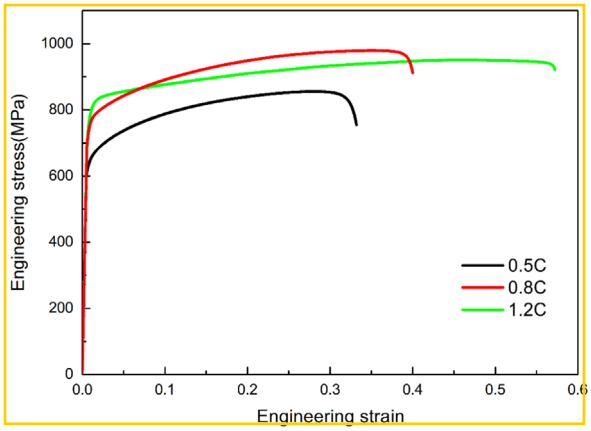

Fig. 3. The engineering stress–strain curves of Fe–18Mn–10Al–*x*C steels.

Table 2
Mechanical properties of 0.5C, 0.8C and 1.2C steels.

Steel	TE/%	YS/MPa	UTS/MPa	UTS × TE/MPa · %
0.5C	32.8(±0.6)	614.9(±1.0)	856.7(±0.7)	28100
0.8C	38.4(±1.6)	711.2(±1.0)	979.1(±1.0)	37597
1.2C	56.2(±1.0)	735.1(±5.0)	946.7(±4.5)	53205

TE: total elongation, YS: yield strength, UTS: ultimate tensile strength.

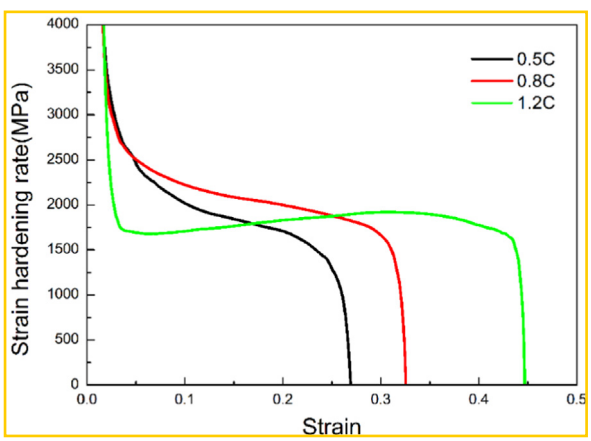

Fig. 4. Variation of strain hardening rate with strain for Fe–18Mn–10Al–*x*C steels.

alloying of the elements even in the quenched state after solution treatment [10]. In the present work, the initial microstructures are quite different in 0.5C and 0.8C steels. Even if the C content increases by 0.3%, CsCl type B2 [(Fe,Mn)Al] and F_{m3m} type DO_3 [$(Fe,Mn)_3Al$] phases appear in ferrite and $L'1_2$ (perovskite) type κ[$(Fe,Mn)_3AlC_x$] in austenite. In 1.2C steel, the matrix is austenitic and κ carbides are uniformly distributed. The existence of these ordered phases must exert effects on the mechanical properties and deformation behavior of the steels.

4.1. Influence of microstructures on mechanical properties

As is observed with increasing C content from 0.5% to 0.8% in Table 2, the yield strength (YS) and UTS of 0.8C steel are much higher than those of 0.5C steel, which mainly results from the combined effects of the ordered phases formed in both ferrite and austenite. It is interesting to notice that the ductility of 0.8C steel is also higher than that of 0.5C steel, which must be associated with the higher and more extended strain hardening rate of the steel, shown in Fig. 4. On the other hand, the YS values of 0.8C and 1.2C steels are similar, though 0.8C steel has a much smaller grain size than 1.2C steel. This may be explained by the fact that a larger fraction of densely populated κ carbides by C addition contributes to an increase in the yield strength of 1.2C steel. The ductility of 1.2C steel is superior to those of other two steels due to the extended strain hardening.

In the present work, 0.8C steel possesses a tensile strength of ~979 MPa and an elongation of ~38.4%, and the product of strength and elongation is as high as 37597 MPa · %, which satisfies the requirement for high performance automobile steels. Its specific strength is 142.3 MPa cm^3/g, which is also quite competitive. The mechanical properties of the alloy are superior to the result of 881 MPa and 38.1% in the Fe–20Mn–10Al–1.2C steel reported in Ref. [7]. The strength and elongation of the present studied steel are also comparable to 995 MPa and 42%, which has been reported in the as-quenched Fe–27Mn–12Al–0.8C steel [10]. Therefore, it can be concluded that the same or a higher level of strength and ductility can be achieved with a lower Mn content or C content by controlled thermo-mechanical processes and subsequent heat treatment schedules.

4.2. Influence of microstructures on strain hardening behavior

As clearly observed in Fig. 4, the microstructural changes have significant effects on the strain hardening rate vs true strain. The 1.2C steel sustains a sufficient strain hardening rate to prevent the accumulation of large strains in local regions after 0.026 strain, while 0.5C and 0.8C steels exhibit the trend of a decreasing strain hardening rate with true strain curves, though the slopes vary to some extent upon straining.

Normally, the tensile stress–strain curves were analyzed by the following several approaches, such as Hollomon analysis, the differential Crussard–Jaoul (C–J) analysis and the modified C–J analysis [15]. So far, strain hardening behavior of dual-phase steels have been investigated extensively [16–18]. It is known that there is an apparent difference in elastic and plastic behavior in soft ferrite and hard martensite, so that strain partitioning between two phases occurs upon loading. Thus, it was stated that Hollomon analysis is not applicable to describe the strain hardening behavior of the ferrite–martenisite dual phase steels [16]. Another research work [17] also demonstrated that Hollomon analysis can predict the initial strain hardening rate in a small strain range, but it cannot reflect the multi-stage strain hardening. Similarly, ferrite and austenite exhibit distinct strain hardening behaviors due to their structural differences, and meanwhile the existence of ordered phases would exert effects on their deformation behaviors. As demonstrated in Section 3.3, as deformation proceeds the complex microstructural changes happen so that a multi-stage strain hardening is expected. Therefore, the C–J analysis has been utilized in this study to analyze the strain hardening behaviors of the investigated steels.

As plotted in Fig. 5, the C–J analysis can describe the strain hardening behaviors of Fe–18Mn–10Al–*x*C steels fairly well. For example, a two-stage strain hardening is observed for 0.5C steel (Fig. 5(a)), which agrees with the microstructural changes, as shown in Fig. 6. In the first stage, planar slip in austenite starts (Fig. 6(a)), whereas dislocation tangles form in ferrite(Fig. 6(b)). In the second stage, a low energy Taylor lattice structure, which is equivalent to the dislocation tangles in materials with wavy slip and microbands develops in austenite (Fig. 6(d) and (e)), and more dislocation tangles form in ferrite (Fig. 6(f)), resulting in a relatively high strain hardening.

It was reported that the C–J analysis was less sensitive to the changes of the ferritic matrix and the shape and distributions of the second phase martensite which were altered by heat treatment [18], therefore, the modified C–J analysis was utilized

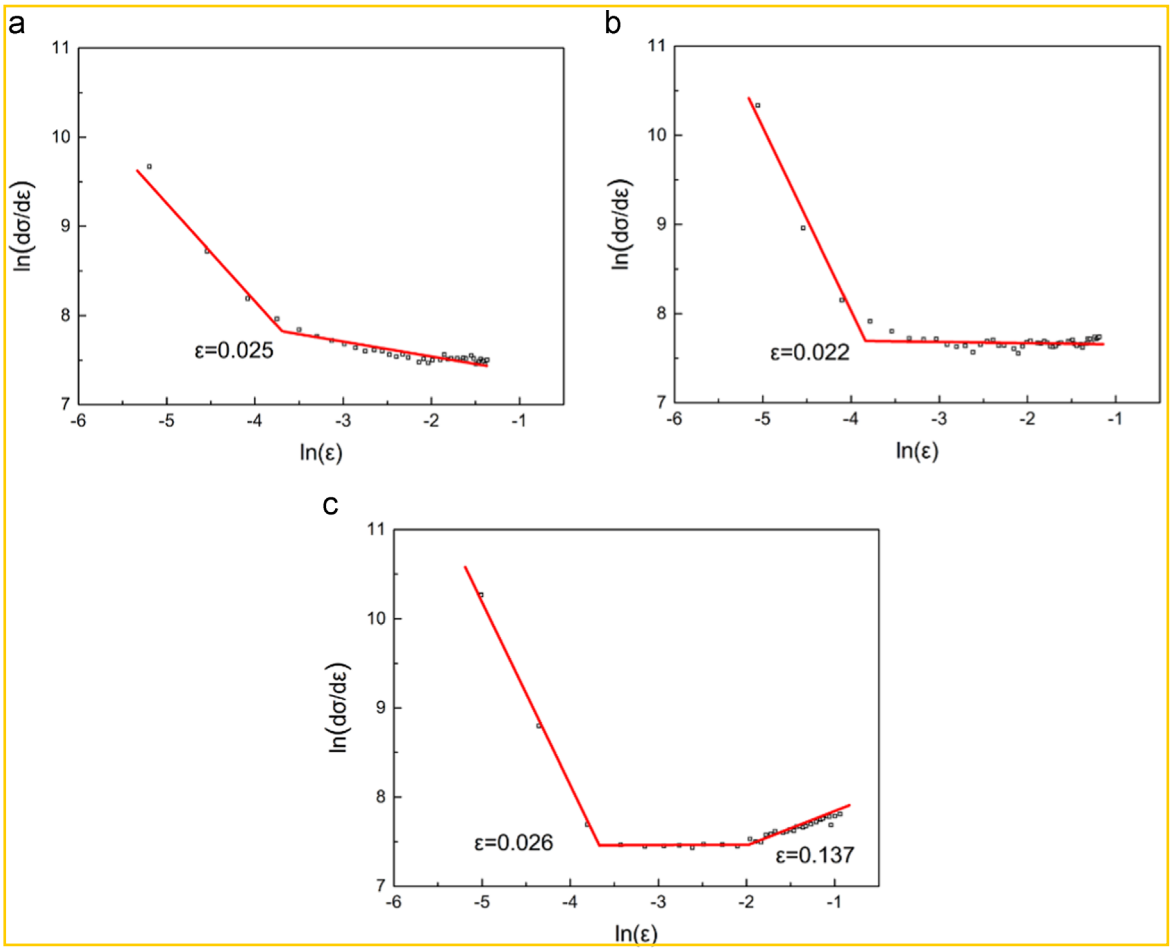

Fig. 5. $\ln(d\sigma/d\varepsilon)$ vs $\ln\varepsilon$ curve for Fe–18Mn–10Al–xC steels (a) 0.5C steel, (b) 0.8C steel, and (c) 1.2C steel.

instead. The modified C–J analysis revealed the duplex steels deform in two stages. The first stage could be associated with the deformation of the ferritic matrix and the second stage was probably induced by the uniform strain of ferrite and martensite. In the present case, the C–J analysis can describe the strain hardening behavior of the duplex Fe–18Mn–10Al–0.5C steel. The reason may be because the features of ferrite and martensite are quite different and the increase in stress in matensite is dominant in strain hardening; whereas the difference in hardness between ferrite and austenite is not quite obvious [8,9] so that the increase in strain can reflect the strain hardening behavior of the 0.5C steel.

A comparison of Fig. 5(a) and (b) indicates that the characteristics of two-stage hardening have not changed with the existence of the ordered precipitates in 0.8C steel (Fig. 5(b)), even though the slope in the first stage is slightly steeper in 0.8C steel than that in 0.5C steel. It was clarified that the strain hardening deteriorates when κ carbides appear in the austenite matrix [19]. Therefore, the slightly faster decline in strain hardening in 0.8C steel than in 0.5C steel is most likely to be attributed to the presence of κ carbides in the austenitic phase. This has also been evidenced by the occurrence of multi-slip in austenite of 0.5C steel (Fig. 6(a)) even at a low strain, but a dominant single slip in austenite of 0.8C steel (Fig. 7(a)) as the presence of the ordered phase makes dislocation multiplication difficult [10].

In the second stage, however, the strain hardening in 0.8C steel is somewhat stronger and also extends to higher strains than that in 0.5C steel, shown in Figs. 4 and 5. This seems to be in contradiction with the aforementioned effect. In our previous work in Fe–12Mn–8Al–0.8C steel which consists of austenite, ferrite and κ carbides, a slowly decreasing rate of strain hardening has been demonstrated in the second stage of deformation by the modified C–J analysis [11]. The modified C–J analysis has also been conducted for the present 0.8C steel, but no obvious descending has been found. Fe–18Mn–10Al–0.8C steel (0.8C steel in the present work) and Fe–12Mn–8Al–0.8C steel consist of similar level of volume fraction of austenite (~80%) and the difference between these two steels is whether ordered phases exist in ferrite. Therefore, it can be deduced that the strengthened ferrite phase resulting from the existence of the ordered phases may affect the strain partition between ferrite and austenite in the investigated 0.8C steel, causing the increasing trend in strain hardening as deformation proceeds.

C–J analysis can describe the sensitivity of strain hardening of materials to the changes of internal lattice defects such as dislocation densities and volume fraction of twins. For fully austenitic TWIP steels, the C–J analysis has been conducted to investigate the strain hardening behavior with respect to the microstructure evolution and the contribution of different deformation mechanisms [20,21]. However, the studies on the stain hardening behavior by using the C–J analysis in high stacking fault energy austenite-based alloys without TWIP effect have not been reported to our best knowledge. In the present work, 1.2C steel exhibits a three-stage strain hardening behavior (Fig. 5(c)), which is different from 0.5C and 0.8C steels.

According to the microstructure analysis during plastic deformation of 1.2C steel, the planar slip develops in the primary deformation stage (Fig. 8(a)), followed by the formation of Taylor lattice (Fig. 8(b)) and microbands (Fig.8(c)) (intermediate stage)

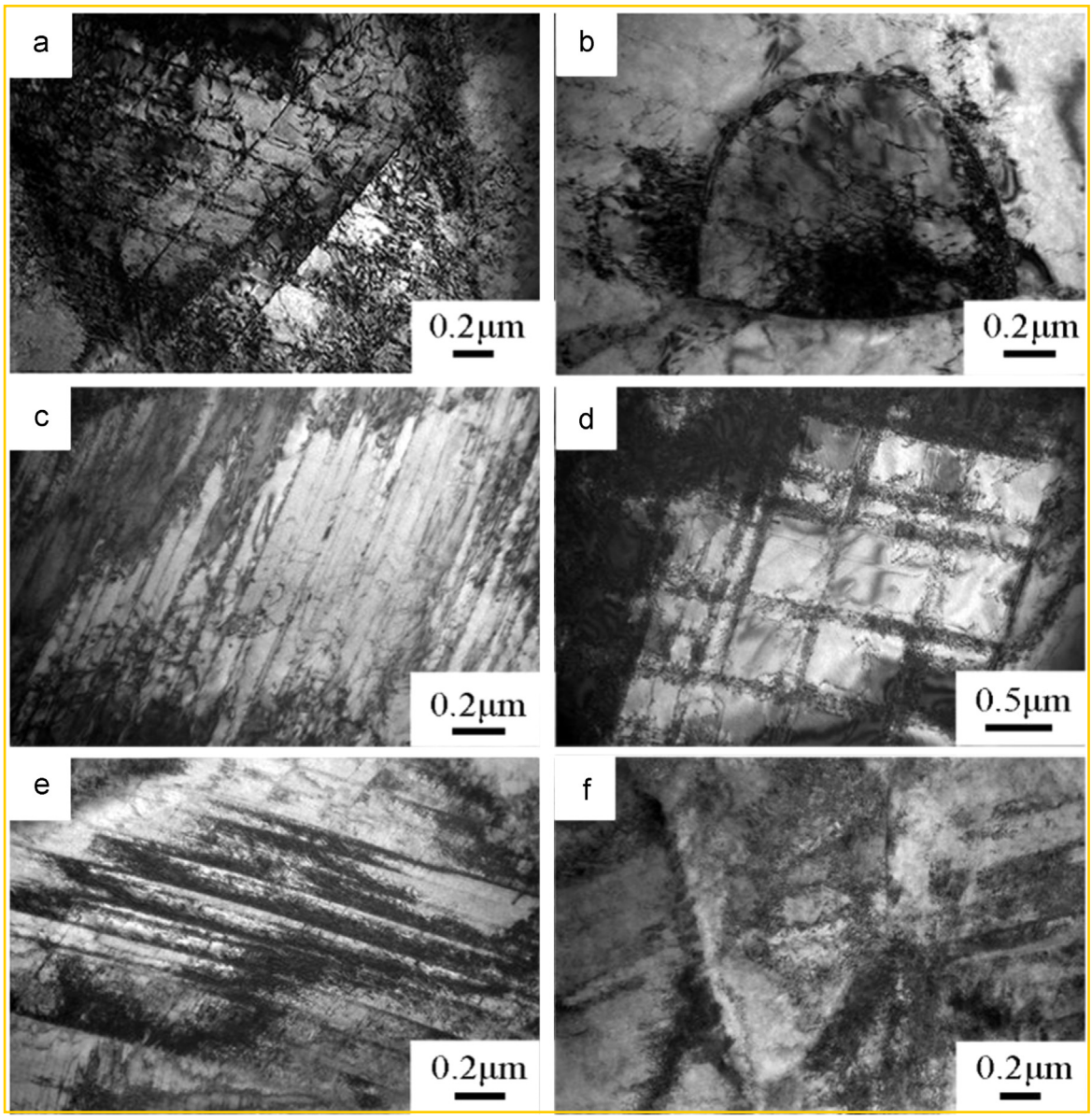

Fig. 6. Microstructural evolution of 0.5C steel (TEM) (a)–(b) $\varepsilon=0.02$: (a) planar slip (γ), (b) dislocation tangles (α), (c)–(f) $\varepsilon=0.18$: (c) intensive dislocation slip (γ), (d) Taylor lattice (γ), (e) microbands (γ), (f) dense dislocation tangles (α).

and the interaction between microbands in the final stage (Fig. 8 (d)). Relating the microstructural evolution to the C–J analysis, it can be deduced that the fast decline of strain hardening in the first stage of 1.2C steel could be mainly related to the easiness of planar slip in the austenite matrix. In the second stage, strain hardening can maintain in large strain regions, which is closely associated with the formation of Taylor lattice and microbands. In the last stage, a continuous increase in strain hardening could be explained by a further increase in the density of microbands and the interaction between microbands. The sustained strain hardening to higher strains enables 1.2 C steel to possess a much higher tensile ductility than 0.5C and 0.8C steels. In contrast, the interaction between microbands has not appeared under TEM due to the limited plastic deformation capability in 0.5C and 0.8C steels, and thus the absence of the 3rd stage leads to a lower tensile ductility in these two steels. In the present work, the C–J analysis can be utilized to describe the multi-stage strain hardening behavior of the austenite based 1.2C steel, demonstrating that such a multi-stage hardening characteristic is a result of the variation of dislocation structures.

From the analysis above, it can be concluded that the C–J analysis can describe the strain hardening behavior of duplex, complex and austenite based low density Fe–Mn–Al–C steels fairly well. Meanwhile, the changes of C content in Fe–Mn–Al steels significantly affect the microstructure-property relationship of Fe–Mn–Al–C steels and the resulting mechanical behavior can be controlled by manipulating the alloying elements.

5. Conclusions

The present work investigates the influence of microstructure differences on strain hardening behaviors during tensile deformation of Fe–18Mn–10Al–xC alloys containing 0.5, 0.8 and 1.2 C (wt%). The main findings can be summarized as follows.

1. The 0.5C and 0.8C steels exhibited a duplex ferritic–austenitic structure, while ordered phases (B2 and DO_3) and κ carbides formed in ferrite and austenite of 0.8C steel, respectively. The 1.2C steel was fully austenitic with some κ carbides.

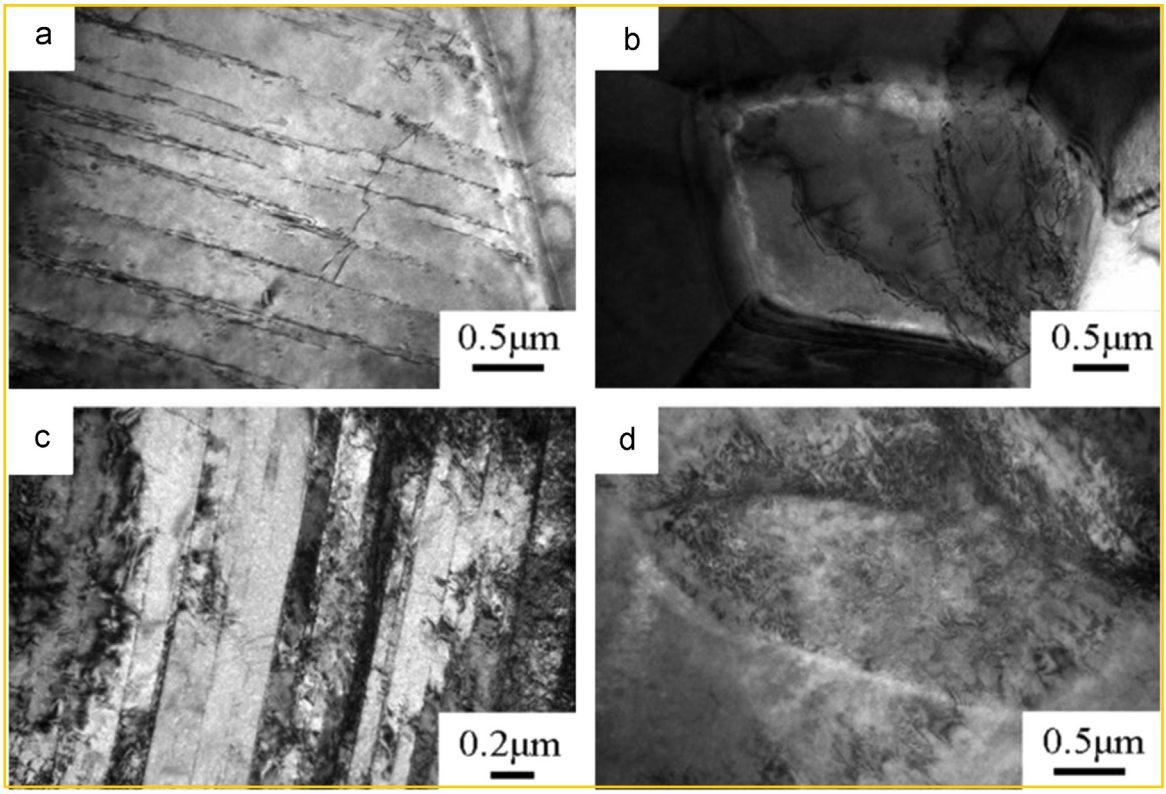

Fig. 7. Microstructural evolution of 0.8C steel (TEM) (a)–(b) $\varepsilon=0.01$: (a) planar slip (γ), (b) dislocations (α), (c)–(d) $\varepsilon=0.18$: (c) microbands (γ), (d) densely populated dislocations (α).

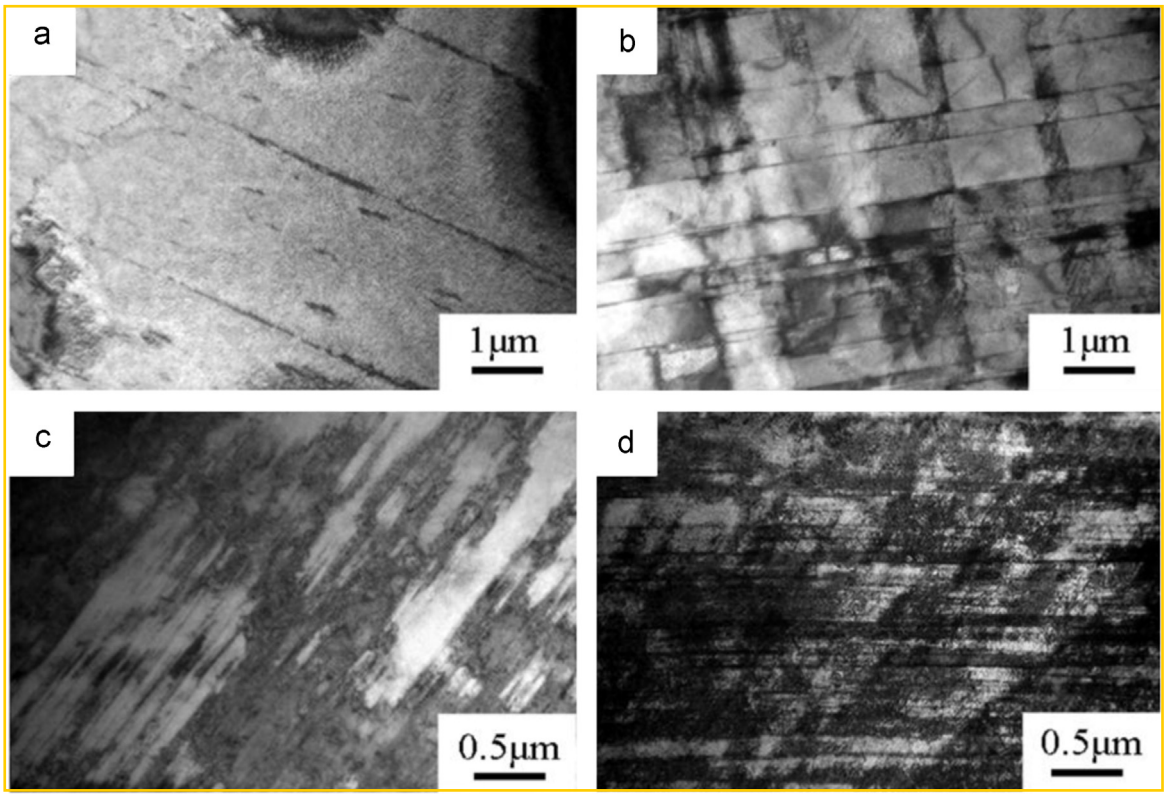

Fig. 8. Microstructural evolution of 1.2C steel (TEM) (a) planar slip ($\varepsilon=0.02$), (b) Taylor lattice ($\varepsilon=0.07$), (c) well developed microbands ($\varepsilon=0.3$), (d) interaction of the microbands ($\varepsilon=0.4$).

2. With increasing C content from 0.5% to 0.8%, the yield strength (YS) and UTS of 0.8C steel are much higher than those of 0.5C steel. Meanwhile, the ductility of 0.8C steel is also higher than that of 0.5C steel. The 0.8C steel exhibited a highest tensile strength of ~979 MPa, and a moderate strain hardening capacity. The 1.2C steel revealed a superior combination of strength

and ductility (ultimate tensile strength of 946.7 MPa and elongation to failure of 56%), with a much more pronounced strain hardening than the other two steels.

3. The C–J analysis can be utilized to evaluate the strain hardening behavior of Fe–18Mn–10Al–*x*C steels properly. It was revealed that 0.5C and 0.8C steels exhibited a two-stage strain hardening, while 1.2C steel manifested a three-stage strain hardening.
4. The hardening mechanisms at different deformation stages were further verified by microstructural changes during tensile testing. The influence of microstructure differences on strain hardening behaviors was well understood.

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